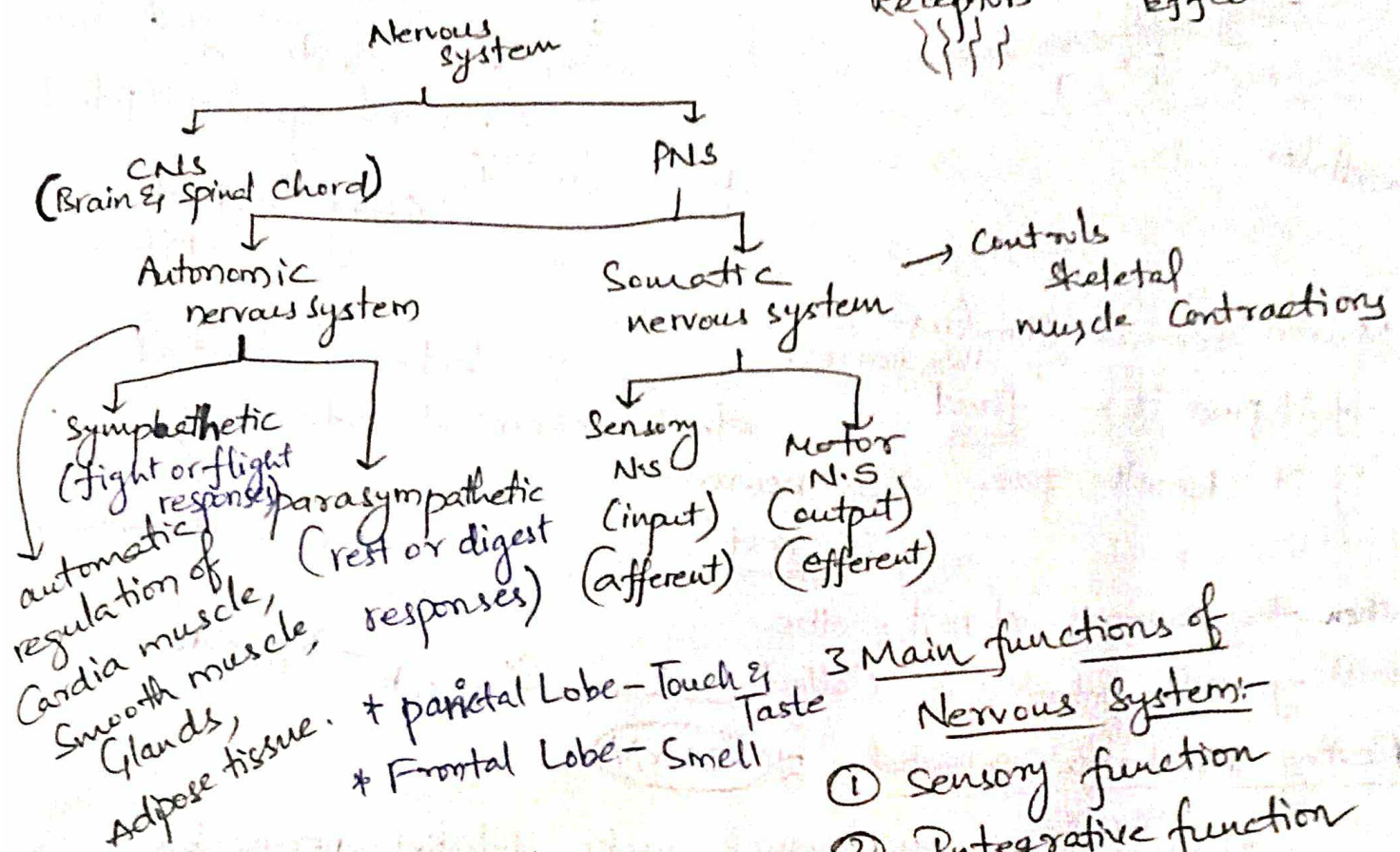


Human Physiology:-



3 Main functions of Nervous System:-

- ① Sensory function
- ② Integrative function
- ③ Motor function

* Structural and functional unit of Nervous System -
Neuron / Nerve cell

* Temporal Lobe - Sound ~~hearing~~ (Ear) ~~noise~~

* Cerebellum - Body balance responsible

* Occipital Lobe - Sight (Eye)

* parietal Lobe - Touch & Taste
* Frontal Lobe - Smell

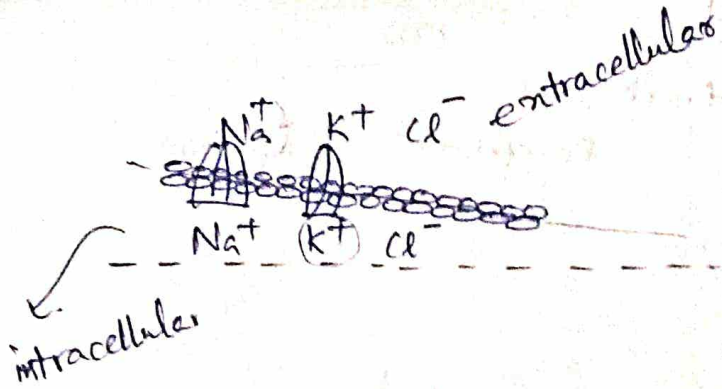
* 12/03/2024

Neurotransmission - passage of info b/w two neurons

Action potential - passage of info within a neuron

Parts of a neuron

- ① Dendrites (Radio antenna)
- ② Axon (Telephone Cable)
- ③ Axon terminals (Telephone Radio transmitter)
- ④ Myelin sheath (acc the flow of nerve impulse)



Na^+ is more outside
 K^+ is more inside

So, there exists a
 Chemical gradient (potential)
 due to concentration $\rightarrow$ leads to diffusion

* Neuron receives stimulus - then neuron is
 if it pass it - "fired"
 if it doesn't pass - then neuron is "rest"

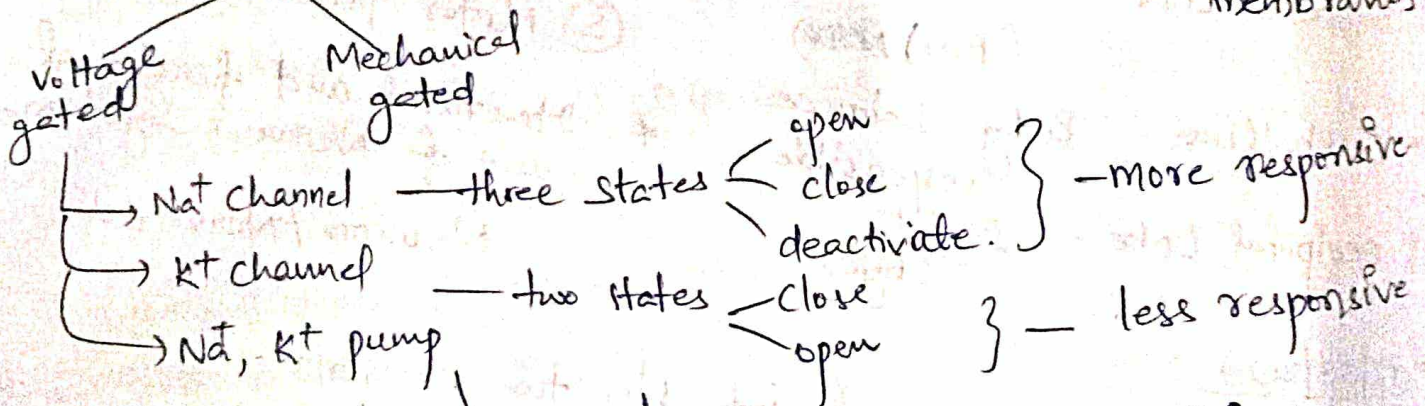
electric potential
 electro-chemical potential
 membrane-potential

* When the neuron is at rest - the existing mem. potential is called

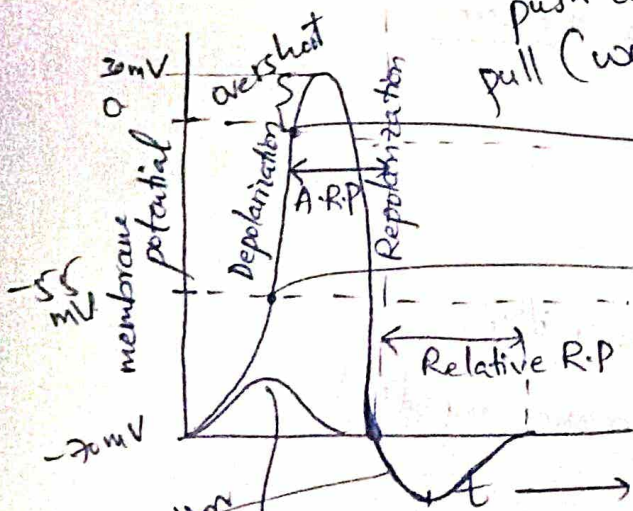
Resting Membrane potential $\approx -70\text{mV}$

$V_{\text{inside}} - V_{\text{outside}}$

* "Channels" - proteins which make selective permeable membranes



push and pull (work both ways) - but we need ATP energy



Na^+ channel deactivates

Na^+ channel (high res) - activate

and Na^+ comes in

Rate of Na^+ Come in $>$ Rate of K^+ go out

hyperpolarization

graded potential which doesn't activate channels

Entire plot is called one ACTION POTENTIAL

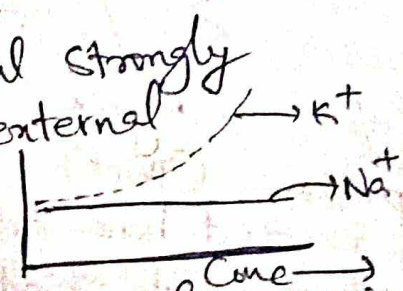
Voltage vs Time — electric signal
(nerve signal)

* In PAIN, freq is high. (— get action potential frequently) — Not change in amplitude

Refractory period — busy with on going thing.

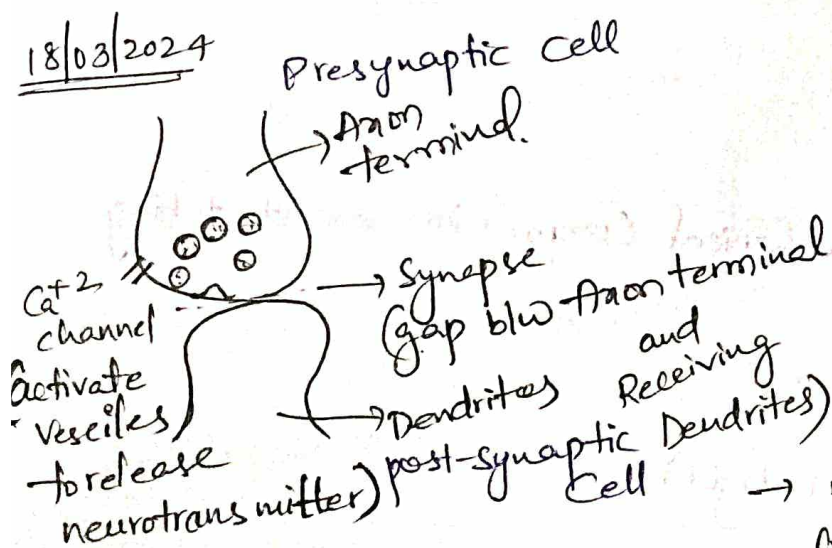
A.R.P R.R.P

* Resting membrane potential strongly depends upon the external K^+ concentration mV



* Synapse — a junction b/w two nerve cells, consisting of a minute gap across which impulses pass by diffusion of a neurotransmitter.

18/03/2024



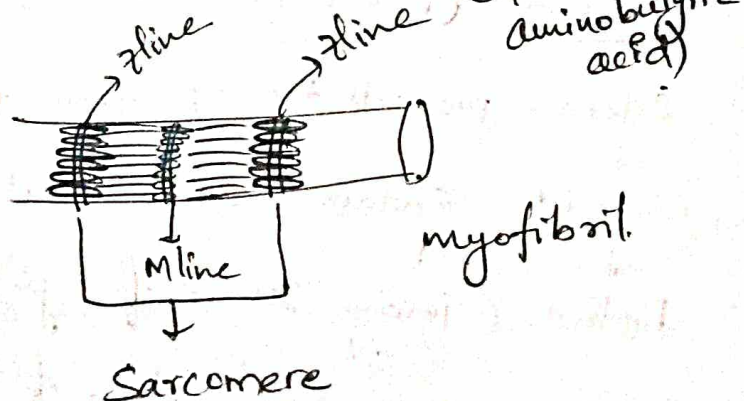
Neurotransmitters
↓
Ext: Glutamate, dopamine

excitatory
→ more likely to fire the next neuron
Ext: Glutamate

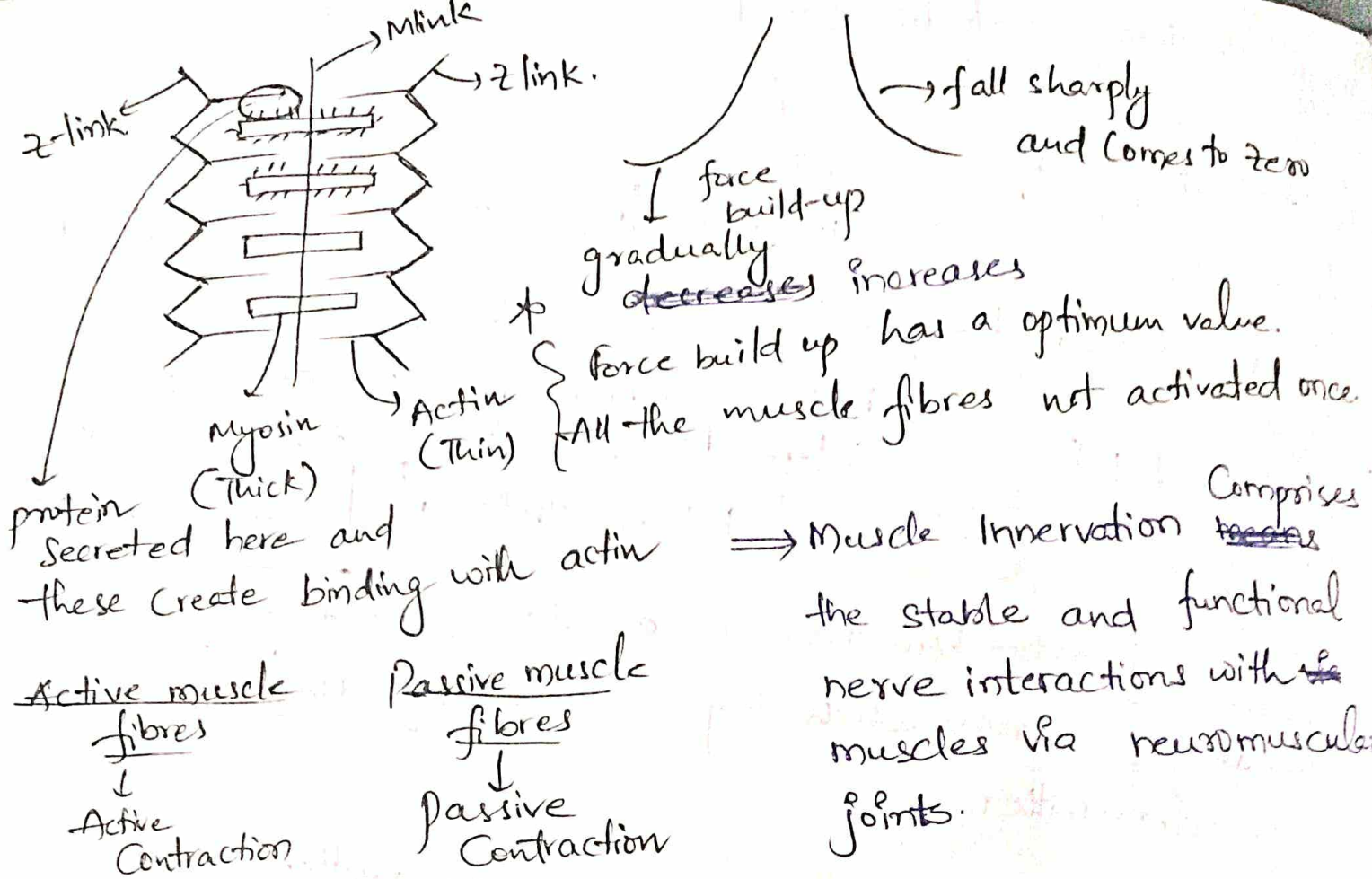
inhibitory
→ less likely to fire the next neuron
Ext: GABA (Gamma-aminobutyric acid)

Skeletal Muscles

Muscle (Epimysium)
↓
fascicle (perimysium)
↓
Muscle fiber (Endomysium)
↓
myofibril
↓
myofilaments
↓
Thick (myosin) Thin (Actin)



myofilaments — Actin, myosin
(long protein molecules) (Thin) (Thick)



22/02/24

Receptors:

- ① Mechanoreceptors - Detect mechanical Compression or stretching
- ② Thermoreceptors
- ③ Nociceptors - Pain receptors
- ④ Electromagnetic receptors (Detect Light)
- ⑤ Chemoreceptors

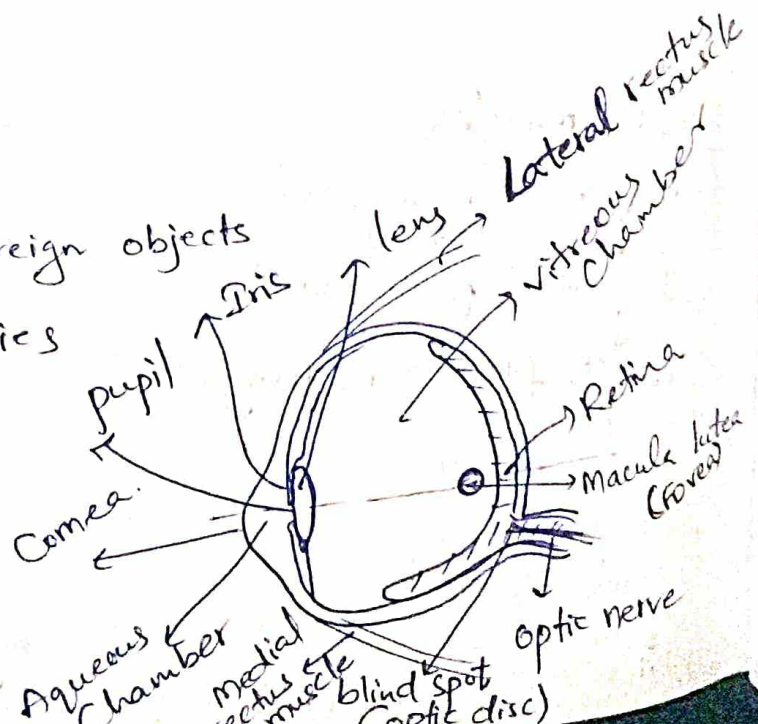
Vision (Eye)

Sclera - protects the eye from foreign objects

↓
Choroid - Contains veins & arteries

↓
Retina (image screen of eye)

Iris - based on which colour of eye is and controls the size of pupil.



macula lutea - brightest spot of eye

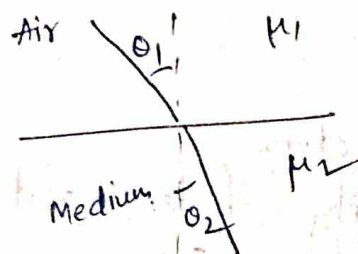
blind spot - doesn't contain photoreceptors
(no image formation)

phototransduction

↓
visual cycle

Rods - Intensity of light detected
Cones - Colour detection

* The lens system of eye is composed of four refractive interfaces
anterior/posterior - front and back surface of the body



$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

$$R.D_m = \frac{\sin \theta_1}{\sin \theta_2} = \frac{\mu_2}{\mu_1} > 1$$

Visible light range - (350-750 nm)

Far object - Iris relaxed -  lens

Near object - Iris contracts -  lens

Bifocal lens -  Concave lens
Convex lens

On retina inverted image of the object is formed.

- concave lens for correction
- Myopia (Nearsightedness)
- Hypermetropia (Farsightedness)
- Convex lens "

Refractive Power of Lens :-
(Diopetre) $D = \frac{1}{f(\text{in m})}$

Convex - +ve $= \frac{100}{f(\text{in cm})}$
Concave - -ve

Emmetropic Vision - Normal Vision (No refractive error or visual defects)
(perfect vision)

27/03/2024

Binocular vision - Humans, Lion - predominant binocular vision
Monocular vision - vision which is made by combination of two eyes

Intraocular fluid - fluid inside the eye
→ for greater depth perception of objects in surroundings

Fluid System

Aqueous humour (Rate of formation: $0.3 \mu\text{l}/\text{min}$)

Vitreous humour

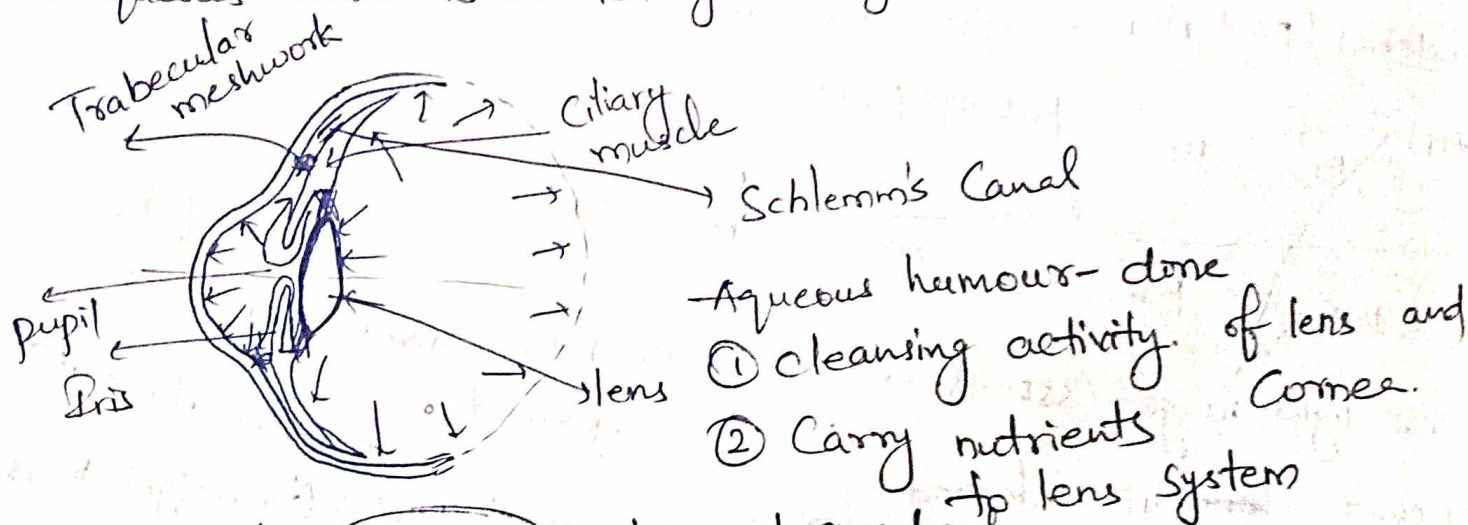
To measure intraocular pressure - (IOP)

Method - Ocular tonometry.

Glaucoma:- Tonometer

→ Important test to evaluate the risk of glaucoma.

→ aqueous humour is secreted by ciliary muscle



open angle glaucoma → slow and gradual

Close angle glaucoma

RGC - Retinal Ganglion Cells

Due to Compression of Optical nerves,

Due to lack of nutrients (due to cytoplasmic stops) ^{flow}

↓
Death of ~~invol~~ fibres (Some nerve cells will die)

Surgical Methods

* photoreceptors work on graded potential

① Laser Trabeculoplasty

② Trabeculectomy - is a type of glaucoma

(because the amplitude matters)
Surgery

for eye for a new pathway for fluid

inside the eye to be drained.

Aqueous humour - free flowing fluid in front of the lens
Vitreous humour - gelatinous mass with little flow of fluid at the back of the lens.

Cataracts = formation of a cloudy area in the lens of our eye.

They make our vision blurry, hazy or less colourful.
It can be treated by a Surgery.

Photoreceptors (Rods and Cones):-

6 million Cones

120 million rods

Rods - responsible for vision at low light levels

Cones - responsible for vision at higher light levels

↳ provide colour vision in bright light

photopsins - (BGR)

Chromophore - The light-catching visual pigments of the retina found in cones

Rhodopsin - The visual pigment found in rods.

Light blindness :- is the trouble seeing in dim or dark settings. It can happen if your eyes have trouble quickly adjusting to changes in brightness.

Causes:-

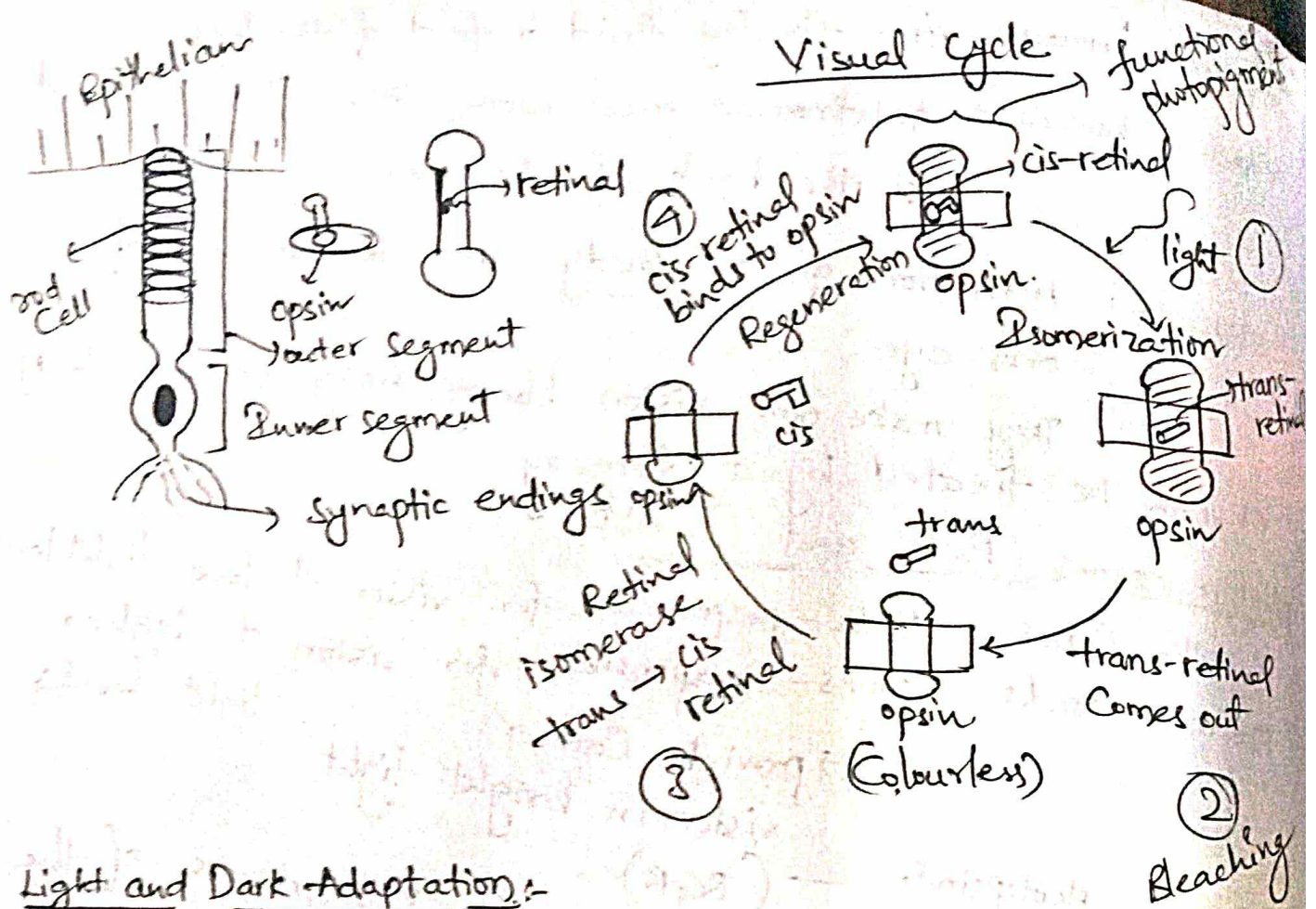
→ Vitamin A deficiency

Visual cycle - Wald's Visual Cycle - Visual phototransduction

photopigment

glycoprotein - opsin

Retinal (Vitamin A derivative)



Light and Dark Adaptation:-

Dark $\rightarrow$ Light Fast: Happens in seconds
 (Cones will play Centre stage)

Light $\rightarrow$ Dark Slow: Happens in few minutes
 (Because rods will play Centre stage)

- ① Photoreceptor cells
- ② ~~Gen~~ Bipolar cells
- ③ Ganglion cells → Generate action potentials

- Horizontal cells $\rightarrow$ detect any changes in intensity of light (both bright and dim light)
- ④ Brain
- * The pigmented layer of retina adjacent to the photoreceptors stores a large quantity of vitamin A and contributes to regeneration process in rods
 - * The extent of rhodopsin generation decreases drastically if the retina detaches from pigmented layer.
 - * Cone pigments regenerate faster than rod pigments.

Hearing

Sound waves are longitudinal waves.

In ear, mechanical energy $\rightarrow$ electrical energy

The ear also contains receptor for equilibrium.

Outer Ear $\rightarrow$ ends at Ear Drum (Tympanic membrane)

Middle Ear

Malleus
Incus
Stapes } 3 bones

Connect via oval window
 $\rightarrow$ Inner ear

Inner Ear

houses receptors
for hearing and
equilibrium

$\hookrightarrow$ there will be
auditory transduction

Outer Ear :-

auricle, external auditory canal, eardrum

perforated ear drum

hearing is not clear
due to small holes on
ear drum

But ear drum - self
repairing
with taken some

impacted cerumen

cerumen solidifies
and attaches to eardrum.

Can be removed by cleaning
via suction with the help of ENT etc.

Middle Precautions.

Inner Ear :-

Auditory ossicles - Malleus, Incus, stapes

Smallest skeletal muscle in our body - Stapedius muscle
(It also amplifies the hearing sound)

$$P_1 A_1 = P_2 A_2$$

$$\frac{P_1}{P_2} = \frac{A_2}{A_1} \quad A_2 \ll A_1 \quad \frac{A_2}{A_1}$$

$$\frac{P_2}{P_1} = \frac{A_1}{A_2} = \frac{55}{3.2} \approx 17 \text{ times}$$

Due to displacement diff
of Malleus and Incus
another 5 times
amplification.
Total ≈ 22 times

Inner Ear

Cochlea

~~Amputi~~

- * Directly below the oval window is another opening called the round window.
- * Amplification is required because the cochlea in the inner ear conducts sound through a liquid, instead of through air. This fluid has much higher inertia than air i.e. it is harder to move.

Inner Ear (Labyrinth):

Bony Labyrinth (Contains perilymph)

Membranous Labyrinth (Contains endolymph)

- * Vestibule houses sensory as well as motor neurons that synapse with receptors for equilibrium.

